

Registration System - Full Testing Plan

Minecraft AI Club Hub | Generated 2026-07-18 | Firebase: website-b84d2

Legend: PASS = expected success FAIL = expected rejection WARN = edge case INFO = reference file = source file L### = line number

Prerequisites

Step	Action	Expected
1	Open terminal in project folder	C:\Users\athit\OneDrive\Desktop\clubsite
2	Run <code>npm run dev</code>	Vite dev server starts on localhost:5173
3	Open two browser tabs: one Admin, one Guest	Both tabs loaded at http://localhost:5173

Test 1 - Guest Registration Saves to Firestore DB

#	Action	Expected Result	File Ref
1	Navigate to /announcements (not logged in)	Announcement list loads PASS	
2	Click any event	Dynamic form fields render from formFields config	AnnouncementDetails.jsx L897-960
3	Fill all required fields, click Register	Form submits via handleRegister()	AnnouncementDetails.jsx L267
4	Watch browser console	Firestore write logged; persistedToFirestore: true returned	db.js L914-974
5	Confirm QR code ticket appears	Guest sees QR + confirmation PASS	
6	Open Firebase Console > Firestore > EventRegistrations	Document visible with announcementId, userEmail, fullName, status: Registered	

Test 2 - Admin Sees Registrations

#	Action	Expected Result	File Ref
1	Log in as admin, go to /admin	Dashboard loads	
2	Go to Announcements section	AnnouncementAdmin.jsx loads	
3	Click View Registrations on Test 1 event	loadRegistrations() runs, Firestore queried	RegistrationDashboard.jsx L18-29
4	Inspect the registration table	Columns visible: Name, Contact, Reg No, Year/Class, Registered On, Status badge PASS	RegistrationDashboard.jsx L420-486
5	Check statistics bar	total, confirmed, waitlisted, checked-in counts match Firestore	RegistrationDashboard.jsx L37-48

Test 3 - Dynamic Form Fields

#	Action	Expected Result	File Ref
1	In admin, create announcement with custom field: select "T-shirt Size" with S/M/L/XL	Announcement saved with custom formFields array	AnnouncementAdmin.jsx L95-103
2	Open that event as guest	Custom "T-shirt Size" dropdown appears in the form PASS	AnnouncementDetails.jsx L923-929
3	Fill all fields including custom, register	Submitted value appears in Firestore doc AND in admin table	

Test 4 - Duplicate Registration Blocked

#	Action	Expected Result	File Ref
1	Register for an event with test1@example.com	Registration succeeds PASS	
2	Register again for SAME event with SAME email	BLOCKED with toast: "Already Registered" FAIL	AnnouncementDetails.jsx L279-286
3	Register with DIFFERENT email (test2@example.com)	Succeeds - no conflict PASS	

Test 5 - Concurrent Registration (No Overselling)

#	Action	Expected Result	File Ref
1	In admin, create event with seatsLimit: 2, waitlistLimit: 1	Event created in Firestore	
2	Register person A	Gets "Registered" (registeredCount = 1) PASS	
3	Register person B	Gets "Registered" (registeredCount = 2, seats full) PASS	
4	Register person C	Gets "Waitlisted" (waitlistedCount = 1, waitlist full) WARN	
5	Register person D	REJECTED: "Event is full. No more seats or waitlist slots available." FAIL	db.js L945-951
6	Stress test: 5+ tabs all register simultaneously for 1-seat event	Only 1 gets Registered, rest Waitlisted/rejected. NO overselling (runTransaction enforced) PASS	db.js L914-974

Test 6 - Cascade Delete (Admin Deletes Announcement)

#	Action	Expected Result	File Ref
1	In admin, note event has 3 registrations in Firestore	Registrations visible in EventRegistrations collection	
2	Click Delete on that announcement	Confirmation dialog: "This will also remove all registrations" INFO	AnnouncementAdmin.jsx L193-207
3	Confirm deletion	All 3 registrations deleted from Firestore, then announcement deleted PASS	AnnouncementAdmin.jsx L199, L201

4	Toast shows	"Announcement and 3 registration(s) removed."	
5	Check Firebase Console	No EventRegistrations docs with that announcementId remain	

Test 7 - Offline Fallback (Only Triggers When Genuine)

#	Action	Expected Result	File Ref
7a	DevTools > Network > set to Offline. Try to register.	Retries 3 times with backoff, then falls back to localStorage. Toast: "Offline Save" PASS	db.js L668-682, L960-973
7b	Go back online, register again	Writes directly to Firestore. NO offline toast PASS	
7c	Break Firestore security rules to deny writes. Register online.	Error THROWN (not swallowed). Console: classifyFirestoreError = permission. NO fallback FAIL	db.js L656-666
7d	Insert doc with undefined values. Register.	Error THROWN. Console: classifyFirestoreError = validation or unknown. NO fallback FAIL	db.js L656-666
7e	Register for a full event (see Test 5 step 5)	Error: "Event is full" THROWN. NO offline toast FAIL	

Test 8 - Error Classification Reference

Error Code	Classification	Falls Back to localStorage?
unavailable / network error	offline	Yes - after 3 retries with exponential backoff (500ms, 1s, 1.5s)
permission-denied	permission	No - error thrown to caller
unauthenticated	auth	No - error thrown to caller
not-found	not-found	No - error thrown to caller
invalid-argument / failed-precondition	validation	No - error thrown to caller
resource-exhausted	quota	No - error thrown to caller
already-exists	duplicate	No - error thrown to caller
anything else	unknown	No - error thrown to caller

Test 9 - Registration Cancellation

#	Action	Expected Result	File Ref
1	As registered user, click Cancel Registration on your ticket	Confirmation prompt appears	
2	Confirm cancellation	Status changes to Cancelled in Firestore. Notification sent PASS	AnnouncementDetails.js L412-419
3	New person registers for the same event	Seat is freed - new registration succeeds PASS	

Quick Smoke Test (2 minutes)

#	Action	Expected
1	<code>npm run dev</code> then open <code>http://localhost:5173</code>	App loads
2	Create event (admin) with seatsLimit: 5	Event appears in list
3	Register as guest (any email)	QR code appears
4	Check Firestore > EventRegistrations	Document exists
5	Check admin RegistrationDashboard	Registration visible in table
6	Delete event (admin)	Registrations cascade-deleted
ALL PASS = System working correctly		